LOTQ-0010 / LINEAR ALGEBRA / BEGINNER / FIELD GUIDE

Linear Algebra Foundations for Quant Research

Vectors, matrices, covariance, regression geometry, PCA, and implementation hazards.

Library of the Quant Educational reference manual

Manual Profile

LIBRARY ID	LOTQ-0010
CATEGORY	Linear Algebra
DIFFICULTY	Beginner
FORMAT	Single-column field-guide manual
USE	Study, lookup, and research-system reference

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Manual map

Each entry is a full-width, single reading path reference card.

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How to Use This Manual

Difficulty: Beginner

Reading Method

- Read the one-sentence meaning first. It gives the mental handle for the concept.
- Use the memory jogger as the recall trigger when you review later.
- Study the quant-use and market-example sections to connect the math to trading, modeling, risk, and backtesting.
- Use the formula block as the compact symbolic form, not as a replacement for understanding.
- Use the implementation note to translate the page into Python, NumPy, pandas, or a research-system habit.

Study Priority

- Master shape: rows, columns, axes, and alignment.
- Master dot products: portfolio returns, linear scores, projections, and factor exposure all reuse the same operation.
- Master covariance matrices: they are the bridge from return data to portfolio risk.
- Master regression geometry: fitted values, residuals, and projections explain many modeling diagnostics.
- Master numerical warnings: rank, conditioning, and broadcasting bugs are practical failure points.

Notebook Practice

- Create tiny arrays by hand and compute dot products, matrix-vector products, covariance, and regression outputs.
- Print shapes after every operation until the rules become automatic.
- Replicate the worked calculations in the appendix using NumPy.
- Force yourself to label what rows and columns mean before writing code.

GUIDE

Notation and Shape Guide

Quant Research Field Guide

Difficulty: Beginner

Core Notation Table

x	Vector, often one observation, one signal, one return series, or one exposure list.
X	Matrix, often rows as observations and columns as features.
w	Portfolio weight vector.
r	Return vector.
R	Return matrix, commonly T dates by N assets.
beta	Coefficient or factor-loading vector.
B	Factor exposure matrix.
Sigma	Covariance matrix.
I	Identity matrix.
A.T	Transpose of matrix A.
A^{-1}	Inverse of A, when it exists.
||x||	Norm or size of vector x.

Shape Rules to Memorize

`X: T x K ; beta: K x 1 ; X beta: T x 1 | w: N x 1 ; r: N x 1 ; w^T r: scalar`

The inner dimensions must match. The output shape comes from the outer dimensions.

Beginner Shape Discipline

- Rows usually represent observations, dates, or asset-date pairs.
- Columns usually represent assets, features, or factors.
- A vector must have a known component order.
- A matrix must have a known row axis and column axis.

GUIDE

Research Pipeline Map

Quant Research Field Guide

Difficulty: Beginner

Pipeline Diagram

Data Pipeline as Matrix Workflow

Linear algebra stores the research workflow as arrays and transformations.

Linear algebra does not replace research judgment. It gives the numerical structure that lets research scale from one asset to many assets and from one feature to many features.

Where Linear Algebra Appears

Data layer	Prices, returns, features, and labels are stored as aligned vectors and matrices.
Model layer	Regression and ML models consume feature matrices and coefficient vectors.
Risk layer	Covariance and correlation matrices describe co-movement and portfolio variance.
Portfolio layer	Weight vectors convert signals into allocations and returns.
Validation layer	Residual vectors, rank checks, and singular values diagnose model stability.

Operating Principle

Every matrix operation must answer three questions: what do the rows mean, what do the columns mean, and what does the output represent?

FOUNDATIONS

Why Linear Algebra Matters in Quant Research

Difficulty: Beginner

One-Sentence Meaning

Linear algebra is the language for storing many market observations, features, assets, and model relationships in compact numerical objects.

Memory Jogger

Quant research is not one number at a time. It is rows, columns, vectors, matrices, and transformations.

Visual Block

Data Pipeline as Matrix Workflow

Linear algebra stores the research workflow as arrays and transformations.

Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Backtests use return vectors, feature matrices, covariance matrices, factor exposure matrices, and model parameter vectors.

Simple Market / Trading Example

A cross-sectional model might store 500 stocks by 30 features in one matrix, then multiply by a coefficient vector to produce 500 scores.

Formula / Notation Intuition

`scores = X beta ; portfolio_return = w^T r`

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

It sits in the data structure layer of the research system: raw data -> aligned arrays -> features -> models -> signals -> portfolios. Implementation idea: Use NumPy arrays or pandas DataFrames, but always track shape, index alignment, and what each axis represents.

Confusions to Avoid + Related Terms

Do not treat linear algebra as abstract decoration. In quant work it is the practical storage and calculation system for high-dimensional information. Related terms: array, vector, matrix, transformation, feature matrix

FOUNDATIONS

Scalars, Vectors, and Matrices

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A scalar is one number, a vector is an ordered list of numbers, and a matrix is a rectangular table of numbers.

Memory Jogger

One value, one row/column of values, or a full grid of values.

Why It Matters in Quant Research

Scalars store single metrics, vectors store one observation across many assets or features, and matrices store many observations across many variables.

Simple Market / Trading Example

A daily portfolio return is a scalar, the returns of 100 stocks today form a vector, and 252 days by 100 stocks forms a return matrix.

Formula / Notation Intuition

```
scalar: r ; vector: r_t ; matrix: R with shape T x N
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is the first typing system for quant data: know whether you are holding one number, one list, or a table.

Python / System Implementation Idea

Print .shape constantly when learning. Shape mistakes are often silent until the output is already wrong.

Confusions to Avoid

A vector is not just any random list. Its order and meaning matter. If the order changes, calculations can become wrong even when the numbers look valid.

Related Terms

scalar, vector, matrix, tensor, dimension

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

FOUNDATIONS

Rows, Columns, and Shape

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Shape describes how many rows and columns an array has, and it determines which operations are valid.

Memory Jogger

Before you compute, ask what the rows mean and what the columns mean.

Why It Matters in Quant Research

Quant datasets usually use rows as time or observations and columns as assets, features, or factors.

Simple Market / Trading Example

A matrix X with shape 1000 x 20 can mean 1000 observations and 20 features. A beta vector of length 20 can map each observation to one score.

Formula / Notation Intuition

```
X: T x K ; beta: K x 1 ; y_hat = X beta gives T x 1
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Shape is the first defense against bad research code because valid math depends on compatible dimensions.

Python / System Implementation Idea

Log shapes after every major transform: raw prices, returns, features, targets, predictions, and weights.

Confusions to Avoid

Do not transpose something just to make code run. First understand whether you are switching observations and variables.

Related Terms

shape, axis, row vector, column vector, transpose

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTORS

Vectors as Feature Lists

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A vector can represent one asset, one time step, one portfolio, or one model state as an ordered list of values.

Memory Jogger

A vector is a compact snapshot.

Why It Matters in Quant Research

Feature vectors are the basic input objects for regression, machine learning, ranking models, and signal construction.

Simple Market / Trading Example

For one stock today, a feature vector might contain momentum, volatility, valuation, liquidity, and earnings surprise values.

Formula / Notation Intuition

```
x_i = [momentum_i, volatility_i, value_i, liquidity_i]
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Feature engineering usually means turning messy market information into clean vectors that can be compared or modeled.

Python / System Implementation Idea

Keep a feature schema file or column list so your training and live signal systems use identical vector ordering.

Confusions to Avoid

The same values in a different order represent a different vector. Labels and ordering are part of the meaning.

Related Terms

feature vector, observation vector, state vector

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTORS

Vector Addition and Subtraction

Quant Research Field Guide

Difficulty: Beginner

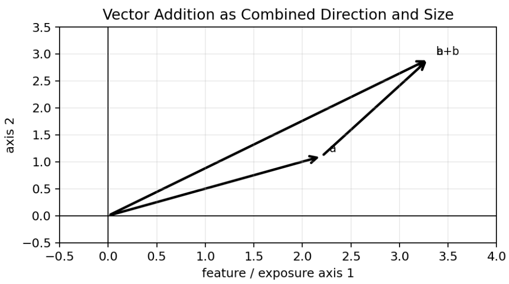
One-Sentence Meaning

Vector addition combines matching components, while subtraction measures component-by-component difference.

Memory Jogger

Add matching slots, not random numbers.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Used for combining exposures, updating positions, computing return differences, and measuring changes in feature states.

Simple Market / Trading Example

If portfolio A has exposures [0.4, 0.1] and portfolio B has [0.2, -0.3], the combined exposure is [0.6, -0.2].

Formula / Notation Intuition

```
a + b = [a1+b1, a2+b2, ..., an+bn]
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

This is the arithmetic layer for changes in multi-asset or multi-feature objects. Implementation idea: Use aligned Series or DataFrames when adding labeled financial vectors to reduce accidental component mismatch.

Confusions to Avoid + Related Terms

Addition only makes sense when components refer to the same axes. Do not add a momentum slot to a volatility slot. Related terms: component, exposure vector, difference vector

VECTORS

Scalar Multiplication

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Scalar multiplication scales every component of a vector by the same number.

Memory Jogger

One knob changes the whole vector proportionally.

Why It Matters in Quant Research

Leverage, position resizing, and exposure scaling are scalar multiplication problems.

Simple Market / Trading Example

If a portfolio weight vector is multiplied by 0.5, every position is cut in half while the relative composition is preserved.

Formula / Notation Intuition

```
c x = [c*x1, c*x2, ..., c*xn]
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is the basic operation behind resizing a signal or portfolio without changing its direction.

Python / System Implementation Idea

Separate signal direction from signal size: direction can come from the vector; size can come from a risk or leverage scalar.

Confusions to Avoid

Scaling preserves direction only for positive scalars. A negative scalar flips the direction.

Related Terms

leverage, normalization, rescaling

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

Dot Product

Difficulty: Beginner

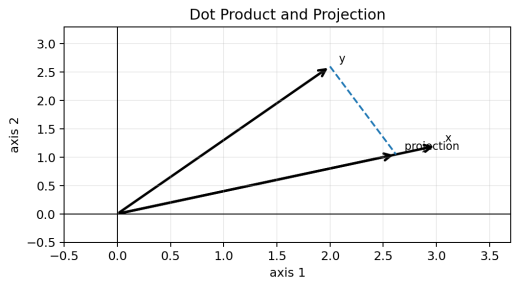
One-Sentence Meaning

The dot product multiplies matching components and adds the results into one scalar.

Memory Jogger

It is a weighted sum and an alignment score at the same time.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Portfolio return, factor exposure, linear model prediction, and projection all use dot products.

Simple Market / Trading Example

If weights are [0.6, 0.4] and asset returns are [2%, -1%], portfolio return is 0.6*2% + 0.4*(-1%).

Formula / Notation Intuition

```
x dot y = sum_i x_i y_i ; y_hat = x dot beta
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

The dot product is one of the most important bridges from vectors to usable scalar decisions. Implementation idea: In NumPy use x @ y or np.dot(x,y). Check that x and y are aligned in the same component order.

Confusions to Avoid + Related Terms

A high dot product can mean large magnitudes, strong alignment, or both. Normalize first if you only want direction similarity. Related terms: inner product, weighted sum, projection, linear score

VECTORS

Norms and Vector Length

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A norm measures the size or magnitude of a vector.

Memory Jogger

A vector can point somewhere; the norm tells how big it is.

Why It Matters in Quant Research

Norms measure exposure size, prediction error, distance from neutral, and residual magnitude.

Simple Market / Trading Example

The L2 norm of residuals summarizes how far model predictions are from actual returns in aggregate.

Formula / Notation Intuition

```
||x||_2 = sqrt(sum_i x_i^2) ; ||x||_1 = sum_i |x_i|
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Norms turn many component errors or exposures into one diagnostic number.

Python / System Implementation Idea

Use norms to monitor portfolio exposure, error size, or drift from a benchmark.

Confusions to Avoid

Different norms emphasize different behavior. L1 emphasizes total absolute amount; L2 penalizes large components more heavily.

Related Terms

L1 norm, L2 norm, Euclidean distance, magnitude

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

Distance Between Vectors

Difficulty: Beginner

One-Sentence Meaning

Distance measures how far apart two vectors are under a chosen norm.

Memory Jogger

Distance is difference plus a size rule.

Why It Matters in Quant Research

Used for nearest-neighbor comparisons, clustering, feature drift detection, and regime similarity.

Simple Market / Trading Example

Today’s market feature vector can be compared with historical days to find similar prior conditions.

Formula / Notation Intuition

```
distance(x,y) = ||x - y||
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Distance metrics help convert high-dimensional market states into comparable objects.

Python / System Implementation Idea

Standardize features before distance-based methods unless the original units intentionally encode importance.

Confusions to Avoid

Raw distance is scale-sensitive. A high-volatility feature can dominate unless variables are standardized.

Related Terms

Euclidean distance, Manhattan distance, feature scaling

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTORS

Unit Vectors and Normalization

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A unit vector has length one and keeps only direction, not magnitude.

Memory Jogger

Normalize when direction matters more than size.

Why It Matters in Quant Research

Useful for comparing signals, exposures, and factor directions without confusing size with alignment.

Simple Market / Trading Example

Two strategy return vectors can be normalized before comparing whether they point in similar return-pattern directions.

Formula / Notation Intuition

```
u = x / ||x||
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Normalization is a preprocessing step that makes comparison and optimization more stable.

Python / System Implementation Idea

Add a small epsilon when normalizing in code to avoid division by zero.

Confusions to Avoid

Do not normalize away meaningful size if magnitude itself carries information, such as liquidity or volatility.

Related Terms

unit vector, scaling, standardization

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTORS

Cosine Similarity

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Cosine similarity measures how aligned two vectors are by comparing their angle rather than their raw size.

Memory Jogger

It asks whether two objects point in the same direction.

Why It Matters in Quant Research

Useful for comparing signals, strategy return profiles, feature vectors, or text/news embeddings.

Simple Market / Trading Example

Two alpha signals may have high cosine similarity even if one has larger amplitude.

Formula / Notation Intuition

```
cos(theta) = (x dot y) / (||x|| ||y||)
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

It belongs in the similarity and redundancy-check layer of research.

Python / System Implementation Idea

Use cosine similarity when direction matters and size should not dominate.

Confusions to Avoid

Cosine similarity ignores magnitude, which can be useful or dangerous depending on the question.

Related Terms

dot product, angle, normalized vectors

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTORS

Orthogonality

Quant Research Field Guide

Difficulty: Beginner

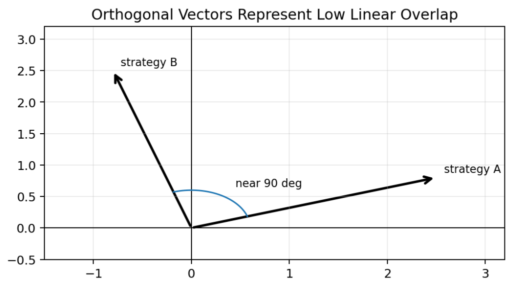
One-Sentence Meaning

Orthogonal vectors have zero dot product, meaning they have no linear overlap under the chosen coordinate system.

Memory Jogger

Ninety-degree directions do not explain each other linearly.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

In quant research, orthogonality is used to discuss independent-looking signals, diversifying return streams, and residualized factors.

Simple Market / Trading Example

A momentum signal residualized against market beta is trying to remove one direction of exposure from another.

Formula / Notation Intuition

`x dot y = 0 means x is orthogonal to y`

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

Orthogonality belongs in signal diagnostics, factor construction, and portfolio diversification. Implementation idea: Check orthogonality with dot products or correlations after standardizing the relevant vectors.

Confusions to Avoid + Related Terms

Orthogonal does not mean truly independent in every sense. It mainly means zero linear relationship under the chosen inner product.
Related terms: dot product, correlation, residualization, independent alpha

VECTOR SPACES

Linear Combinations

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A linear combination is a weighted sum of vectors.

Memory Jogger

Mix building blocks with weights.

Why It Matters in Quant Research

Portfolios, factor models, and regression predictions are all linear combinations.

Simple Market / Trading Example

A portfolio return is a linear combination of asset returns using portfolio weights.

Formula / Notation Intuition

```
v = a1*v1 + a2*v2 + ... + ak*vk
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Linear combinations are the central construction rule for linear models and portfolios.

Python / System Implementation Idea

Treat weights as first-class objects. Store them, audit them, and test how outputs change when they move.

Confusions to Avoid

The weights determine the mix. Bad weights can create leverage, hidden concentration, or unintended exposures.

Related Terms

weights, portfolio, factor model, basis

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTOR SPACES

Span

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

The span of a set of vectors is every vector you can create from their linear combinations.

Memory Jogger

Span is the reachable space of your building blocks.

Why It Matters in Quant Research

In modeling, span tells what kinds of outcomes your chosen features or factors can represent.

Simple Market / Trading Example

If your factor set spans only market beta and value, it cannot represent a pure momentum direction well.

Formula / Notation Intuition

```
span(v1,...,vk) = {a1*v1 + ... + ak*vk}
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Span helps explain model capacity in linear terms.

Python / System Implementation Idea

Use rank checks to see whether new features add independent directions or mostly duplicate existing ones.

Confusions to Avoid

Adding more vectors does not always expand span if the new vector is redundant with existing ones.

Related Terms

basis, rank, column space, redundancy

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTOR SPACES

Basis

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A basis is a minimal set of independent vectors that can represent every vector in a space.

Memory Jogger

A coordinate system needs clean building blocks.

Why It Matters in Quant Research

Bases appear in factor models, PCA, eigenvector decompositions, and coordinate transformations.

Simple Market / Trading Example

Principal components form a new basis that explains variation in descending order.

Formula / Notation Intuition

```
Any vector x can be represented as weighted basis vectors
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Basis thinking helps you understand why the same data can be represented in different coordinate systems.

Python / System Implementation Idea

When reducing dimensions, ask which basis directions preserve the most useful information.

Confusions to Avoid

A basis must both span the space and avoid redundancy. Too many dependent vectors are not a clean basis.

Related Terms

dimension, independence, PCA, coordinates

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTOR SPACES

Dimension

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Dimension is the number of independent directions needed to describe a space.

Memory Jogger

Dimension counts independent degrees of freedom.

Why It Matters in Quant Research

High-dimensional data appears everywhere in quant research: many assets, many features, many time steps, many regimes.

Simple Market / Trading Example

A 500-stock universe has a 500-dimensional return vector at each time step, before any factor compression.

Formula / Notation Intuition

```
dimension = number of independent basis directions
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Dimension controls complexity, sample-size needs, and overfitting risk.

Python / System Implementation Idea

Track the number of observations versus the number of features. Too many features for too little data is a warning sign.

Confusions to Avoid

More dimensions do not automatically mean more information. They may add noise, redundancy, or instability.

Related Terms

rank, basis, feature count, curse of dimensionality

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

MATRICES

Matrices as Data Tables

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A matrix is a rectangular numerical object that can store observations by variables, assets by factors, or time by instruments.

Memory Jogger

Rows and columns turn messy data into computable structure.

Why It Matters in Quant Research

Almost every serious quant pipeline eventually becomes matrix operations.

Simple Market / Trading Example

A return matrix R can have rows as dates and columns as assets.

Formula / Notation Intuition

```
R has shape T x N: T dates, N assets
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Matrices are the main storage format for batch computation and model fitting.

Python / System Implementation Idea

Keep index alignment explicit when moving between pandas and NumPy, because NumPy arrays do not carry labels.

Confusions to Avoid

A matrix with no clear row and column meaning is dangerous. Always name the axes conceptually.

Related Terms

DataFrame, array, design matrix, return matrix

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

MATRICES

Matrix Shape Rules

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Matrix operations require compatible inner dimensions and produce predictable output shapes.

Memory Jogger

The middle dimensions must match.

Why It Matters in Quant Research

Shape rules prevent invalid portfolio calculations, regression fits, and factor exposure operations.

Simple Market / Trading Example

If X is T x K and beta is K x 1, X beta is T x 1 predictions.

Formula / Notation Intuition

`(m x n) times (n x p) = (m x p)`

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is the mechanical grammar of matrix workflows.

Python / System Implementation Idea

Write expected shapes in comments next to important operations until the pattern becomes automatic.

Confusions to Avoid

Broadcasting can make wrong calculations run without error. Valid code is not always valid math.

Related Terms

dimensions, broadcasting, transpose, matrix multiplication

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

MATRICES

Matrix-Vector Multiplication

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Matrix-vector multiplication applies each row of a matrix to a vector, usually producing one score per row.

Memory Jogger

Rows meet weights and produce outputs.

Why It Matters in Quant Research

Used in linear models, portfolio returns through time, factor scoring, and transformations.

Simple Market / Trading Example

A feature matrix times a coefficient vector produces a prediction for every observation.

Formula / Notation Intuition

```
y_hat = X beta
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This operation is the workhorse of batch linear prediction.

Python / System Implementation Idea

Before multiplying, verify that X.columns matches beta.index in the same order.

Confusions to Avoid

If rows and coefficients are misaligned, the output may look plausible but represent the wrong model.

Related Terms

dot product, prediction vector, coefficient vector

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

Matrix-Matrix Multiplication

Difficulty: Beginner

One-Sentence Meaning

Matrix-matrix multiplication combines rows from the left matrix with columns from the right matrix through dot products.

Memory Jogger

A grid of dot products becomes a new matrix.

Why It Matters in Quant Research

Used in covariance construction, factor returns, projections, transformations, and batch scoring.

Simple Market / Trading Example

Asset returns by factors can be created or decomposed using exposure matrices and factor-return matrices.

Formula / Notation Intuition

```
C_ij = row_i(A) dot col_j(B)
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This operation lets a whole system of relationships be computed at once.

Python / System Implementation Idea

Use @ for matrix multiplication in Python and * only when you intentionally want elementwise multiplication.

Confusions to Avoid

Elementwise multiplication and matrix multiplication are different operations. Confusing them is a common beginner bug.

Related Terms

dot product, transformation, composition, matrix product

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

MATRICES

Linear Transformations

Quant Research Field Guide

Difficulty: Beginner

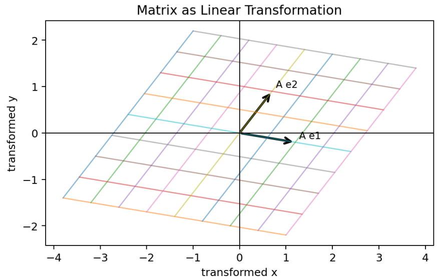
One-Sentence Meaning

A linear transformation maps vectors to new vectors while preserving addition and scalar multiplication structure.

Memory Jogger

A matrix is a machine that moves, stretches, rotates, or mixes coordinates.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Used for factor rotations, PCA, risk transformations, and converting raw features into model-ready representations.

Simple Market / Trading Example

A PCA transform maps correlated features into uncorrelated component coordinates.

Formula / Notation Intuition

```
T(x) = A x
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

This concept explains what matrices do geometrically, not just numerically. Implementation idea: Visualize simple 2D transformations to build intuition before applying high-dimensional transforms.

Confusions to Avoid + Related Terms

Not every transformation is linear. Ranking, clipping, thresholds, and nonlinear features break linearity. Related terms: matrix, basis change, rotation, scaling

Identity and Zero Matrices

Difficulty: Beginner

One-Sentence Meaning

The identity matrix leaves vectors unchanged, while the zero matrix maps everything to zero.

Memory Jogger

Identity means do nothing; zero means erase.

Why It Matters in Quant Research

Identity matrices appear in regularization, covariance adjustments, and numerical stabilization.

Simple Market / Trading Example

Adding a small amount of identity to a covariance matrix can stabilize inversion.

Formula / Notation Intuition

```
I x = x ; 0 x = 0
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

These are baseline matrix objects used in many algorithms.

Python / System Implementation Idea

Use np.eye(n) for identity matrices and np.zeros(shape) for zero matrices.

Confusions to Avoid

The identity matrix must be square. A zero matrix can have many shapes.

Related Terms

identity, diagonal matrix, regularization, shrinkage

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

MATRICES

Transpose

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

The transpose flips rows and columns of a matrix.

Memory Jogger

Rows become columns; columns become rows.

Why It Matters in Quant Research

Transposes appear in dot products, covariance formulas, regression, and portfolio variance.

Simple Market / Trading Example

A return matrix R with shape T x N can be transposed to N x T when comparing assets across time.

Formula / Notation Intuition

```
(A^T)_ij = A_ji
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Transpose is a shape and perspective change.

Python / System Implementation Idea

In Python use A.T, then immediately check the resulting shape and interpretation.

Confusions to Avoid

Do not transpose to fix errors blindly. Make sure the new orientation matches the meaning of the calculation.

Related Terms

row vector, column vector, covariance, regression

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

MATRICES

Inverse Matrices

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

An inverse matrix reverses the action of a square matrix when that reversal is possible.

Memory Jogger

It is the undo button for a linear transformation.

Why It Matters in Quant Research

Inverses appear in solving equations, regression formulas, and portfolio optimization, though direct inversion is often avoided in production.

Simple Market / Trading Example

If A maps inputs to outputs, A^{-1} maps those outputs back to inputs, assuming A is invertible.

Formula / Notation Intuition

```
A^{-1} A = I
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This explains why rank and conditioning matter before solving systems.

Python / System Implementation Idea

Prefer np.linalg.solve(A,b) over computing np.linalg.inv(A) @ b when solving systems.

Confusions to Avoid

Not every square matrix has an inverse. Nearly singular matrices can behave badly even if an inverse technically exists.

Related Terms

singular matrix, rank, conditioning, solve

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

LINEAR SYSTEMS

Systems of Linear Equations

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A system of linear equations asks for unknown values that satisfy several linear relationships at once.

Memory Jogger

Many constraints, one consistent solution if the structure allows it.

Why It Matters in Quant Research

Used in hedging, calibration, factor replication, and solving constrained exposures.

Simple Market / Trading Example

To hedge two factor exposures, solve for instrument weights that offset both exposures simultaneously.

Formula / Notation Intuition

$$A x = b$$

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Linear systems convert model constraints into calculable weights or parameters.

Python / System Implementation Idea

Use linear solve routines and inspect residuals to confirm the solution actually satisfies the equations.

Confusions to Avoid

A system can have one solution, no solution, or infinitely many solutions depending on rank and consistency.

Related Terms

Ax=b, inverse, rank, constraints

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

Rank

Difficulty: Beginner

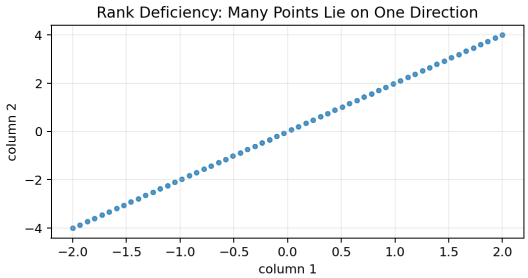
One-Sentence Meaning

Rank measures the number of independent rows or columns in a matrix.

Memory Jogger

Rank counts useful independent directions.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Rank tells whether features are redundant, systems are solvable, and covariance matrices are stable enough to use.

Simple Market / Trading Example

If two features are almost identical, adding both may not add much rank but can increase instability.

Formula / Notation Intuition

```
rank(A) = number of independent columns of A
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

Rank is a structural diagnostic for data and models. Implementation idea: Use np.linalg.matrix_rank(A) as an initial check, but remember numeric rank depends on tolerance.

Confusions to Avoid + Related Terms

High column count is not the same as high rank. Redundant columns can make a large matrix effectively low-dimensional. Related terms: independence, span, multicollinearity, singular matrix

LINEAR SYSTEMS

Full Rank vs Rank Deficiency

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A full-rank matrix has as much independent information as its shape allows, while a rank-deficient matrix contains redundancy.

Memory Jogger

Full rank means no hidden duplicate direction.

Why It Matters in Quant Research

Rank deficiency warns that regression, optimization, or inversion may be unstable or impossible.

Simple Market / Trading Example

A design matrix with two identical features is rank deficient.

Formula / Notation Intuition

```
full column rank: rank(X) = number of columns
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is a core diagnostic before trusting fitted coefficients or optimized weights.

Python / System Implementation Idea

Check correlations, singular values, and condition numbers before fitting large linear models.

Confusions to Avoid

Rank deficiency can be exact or approximate. Approximate redundancy is often the real market problem.

Related Terms

rank, collinearity, conditioning, regularization

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

Column Space

Difficulty: Beginner

One-Sentence Meaning

The column space of a matrix is the set of all vectors that can be created from linear combinations of its columns.

Memory Jogger

It is what the matrix can actually explain or reach.

Why It Matters in Quant Research

In regression, the fitted values live in the column space of the design matrix.

Simple Market / Trading Example

If the target return pattern is outside the feature column space, a linear model can only approximate it.

Formula / Notation Intuition

```
Col(A) = span of columns of A
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Column space explains model capacity and projection geometry.

Python / System Implementation Idea

Think of features as directions the model can use to explain the target.

Confusions to Avoid

Adding weak or redundant columns may not meaningfully expand the column space.

Related Terms

span, basis, projection, least squares

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

VECTOR SPACES

Null Space

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

The null space contains vectors that a matrix maps to zero.

Memory Jogger

These directions disappear under the transformation.

Why It Matters in Quant Research

Null spaces appear in hedging, constraints, and redundant portfolios.

Simple Market / Trading Example

A portfolio in the null space of a factor exposure matrix has zero exposure to those factors.

Formula / Notation Intuition

```
Null(A) = {x : A x = 0}
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Null space is useful for finding portfolios or trades that satisfy neutralization constraints.

Python / System Implementation Idea

Use null-space routines when constructing factor-neutral portfolios or constraint-preserving adjustments.

Confusions to Avoid

Zero exposure to modeled factors does not mean zero risk. It only means zero under that specific matrix model.

Related Terms

kernel, constraints, neutral portfolio, rank

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

LINEAR SYSTEMS

Determinants

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A determinant is a scalar summary related to how a square matrix scales volume and whether it is invertible.

Memory Jogger

If the determinant is zero, the transformation collapses space.

Why It Matters in Quant Research

Determinants are less common in day-to-day quant code but useful for understanding invertibility and transformations.

Simple Market / Trading Example

A zero determinant means a 2D transformation may squash the plane onto a line, losing information.

Formula / Notation Intuition

```
det(A) = 0 implies A is singular
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This concept supports intuition for singularity and rank.

Python / System Implementation Idea

For real workflows, inspect condition numbers or singular values rather than relying only on determinant values.

Confusions to Avoid

Do not overuse determinants as numerical diagnostics in large systems; condition numbers and singular values are usually more useful.

Related Terms

invertibility, rank, volume scaling, singular matrix

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

DECOMPOSITIONS

Eigenvectors and Eigenvalues

Quant Research Field Guide

Difficulty: Beginner

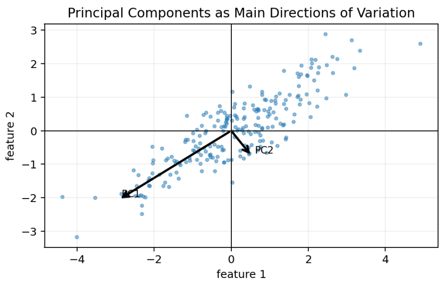
One-Sentence Meaning

An eigenvector is a direction that a matrix only stretches or shrinks, and the eigenvalue is the stretch factor.

Memory Jogger

Some directions survive a transformation without turning.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Eigenvectors explain PCA, covariance structure, factor rotations, and stability modes.

Simple Market / Trading Example

In a covariance matrix, large eigenvalues identify directions where returns vary the most.

Formula / Notation Intuition

```
A v = lambda v
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

Eigen-decomposition turns a complicated matrix into interpretable directions and strengths. Implementation idea: Use eigen-analysis to inspect dominant risk directions, but test stability across time windows.

Confusions to Avoid + Related Terms

Eigenvectors depend on the matrix and scaling. Do not treat them as permanent truths about the market. Related terms: PCA, covariance matrix, principal component, eigenvalue

DECOMPOSITIONS

Diagonal Matrices and Diagonalization

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A diagonal matrix has nonzero values only on the main diagonal, making many calculations simpler.

Memory Jogger

Diagonal means each coordinate scales separately.

Why It Matters in Quant Research

Diagonalization represents a matrix in a coordinate system where the main action is simple scaling along special directions.

Simple Market / Trading Example

A covariance matrix can be decomposed into principal directions and variances along those directions.

Formula / Notation Intuition

```
A = P D P^{-1} when diagonalization is possible
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This supports PCA and risk decomposition intuition.

Python / System Implementation Idea

Use library decomposition routines; do not implement eigenvalue algorithms by hand while learning.

Confusions to Avoid

Not every matrix is nicely diagonalizable, and numerical issues can make decompositions unstable.

Related Terms

eigenvectors, eigenvalues, PCA, basis change

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

Covariance Matrices

Difficulty: Beginner

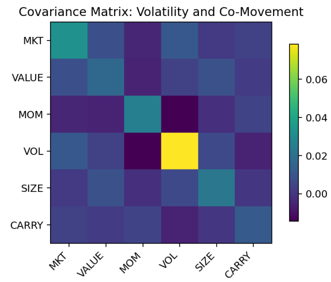
One-Sentence Meaning

A covariance matrix stores variances on the diagonal and pairwise co-movements off the diagonal.

Memory Jogger

It is the table of how risks move together.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Covariance matrices drive portfolio risk, diversification analysis, risk models, and optimization.

Simple Market / Trading Example

If two assets have high positive covariance, holding both may not diversify as much as it first appears.

Formula / Notation Intuition

```
Sigma_ij = Cov(r_i, r_j)
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

This is one of the most important matrices in quant finance. Implementation idea: Estimate covariance with enough data, then consider shrinkage or factor models for stability.

Confusions to Avoid + Related Terms

Covariance estimates are noisy, especially with many assets and limited history. Related terms: variance, correlation, portfolio risk, Sigma

RISK AND PORTFOLIOS

Correlation Matrices

Quant Research Field Guide

Difficulty: Beginner

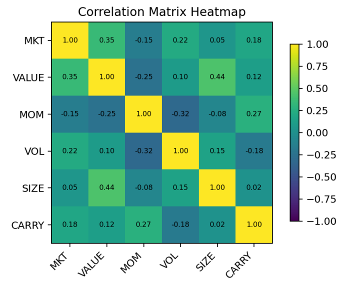
One-Sentence Meaning

A correlation matrix standardizes covariance into values between -1 and 1 that describe linear co-movement strength.

Memory Jogger

Correlation removes volatility scale and focuses on direction of co-movement.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Useful for redundancy checks, diversification analysis, and feature dependence diagnostics.

Simple Market / Trading Example

Two signals with correlation 0.95 likely contain very similar linear information.

Formula / Notation Intuition

```
corr(X,Y) = Cov(X,Y) / (std(X) std(Y))
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

Correlation matrices help detect crowded or duplicate ideas. Implementation idea: Compute rolling correlations to see whether relationships are stable or regime-dependent.

Confusions to Avoid + Related Terms

Correlation does not prove causation and may change sharply across market regimes. Related terms: covariance, diversification, redundancy, regime change

RISK AND PORTFOLIOS

Positive Semidefinite Matrices

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A positive semidefinite matrix never produces negative quadratic variance for any vector.

Memory Jogger

Risk cannot be negative for any portfolio direction.

Why It Matters in Quant Research

Covariance matrices should be positive semidefinite because portfolio variance must be nonnegative.

Simple Market / Trading Example

If a covariance matrix gives negative variance for some weights, something is broken numerically or structurally.

Formula / Notation Intuition

```
x^T A x >= 0 for all x
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This property is a sanity check for risk models and optimizers.

Python / System Implementation Idea

Check eigenvalues of covariance estimates. Small negative eigenvalues may require cleaning or regularization.

Confusions to Avoid

Empirical covariance matrices can become unstable or nearly singular even when theoretically valid.

Related Terms

quadratic form, covariance, eigenvalues, portfolio variance

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

RISK AND PORTFOLIOS

Portfolio Weights as a Vector

Quant Research Field Guide

Difficulty: Beginner

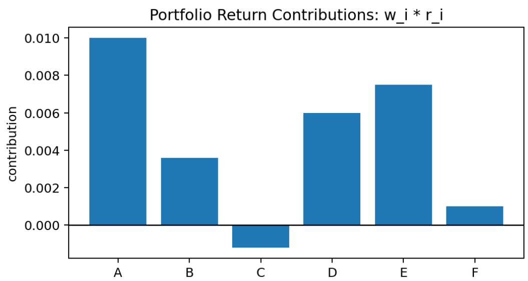
One-Sentence Meaning

A portfolio weight vector stores how capital is allocated across assets.

Memory Jogger

Weights are the portfolio written as math.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Portfolio returns, exposures, risk, and constraints all start with the weight vector.

Simple Market / Trading Example

If w is weights and r is asset returns, portfolio return is w^T r.

Formula / Notation Intuition

```
R_p = w^T r = sum_i w_i r_i
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

Weight vectors are the decision variables of portfolio construction. Implementation idea: Store weights with asset identifiers and rebalance timestamps. Never rely only on raw array position.

Confusions to Avoid + Related Terms

Weights can hide leverage if long and short positions offset in net terms. Track gross exposure too. Related terms: portfolio return, exposure, leverage, constraints

RISK AND PORTFOLIOS

Portfolio Variance Formula

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Portfolio variance is a quadratic form combining weights and the covariance matrix.

Memory Jogger

Risk is weights, co-movement, and scale all interacting.

Why It Matters in Quant Research

This formula is the backbone of mean-variance optimization and risk attribution.

Simple Market / Trading Example

A small position in a highly volatile or highly correlated asset can still contribute a lot of portfolio variance.

Formula / Notation Intuition

```
Var(R_p) = w^T Sigma w
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This page connects vectors, matrices, and portfolio risk in one expression.

Python / System Implementation Idea

Use this formula directly in risk checks and compare it to realized portfolio variance as validation.

Confusions to Avoid

Portfolio variance is not just a weighted average of individual variances because covariances matter.

Related Terms

covariance matrix, quadratic form, risk model

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

FACTOR MODELS

Factor Exposure Matrix

Quant Research Field Guide

Difficulty: Beginner

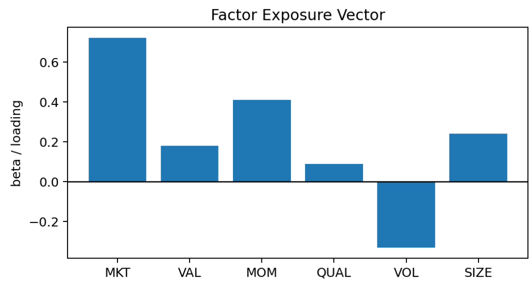
One-Sentence Meaning

A factor exposure matrix stores how sensitive each asset or portfolio is to each factor.

Memory Jogger

Rows are assets, columns are factor loadings.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Factor models compress many asset risks into fewer explanatory directions such as market, value, momentum, size, and volatility.

Simple Market / Trading Example

A portfolio’s factor exposure can be computed by multiplying asset weights by the exposure matrix.

Formula / Notation Intuition

```
portfolio_factor_exposure = w^T B
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

This is central to understanding hidden bets in a portfolio. Implementation idea: Audit factor exposures before and after optimization to verify the portfolio is taking the intended bets.

Confusions to Avoid + Related Terms

Factor labels do not guarantee clean economic meaning. Exposures depend on how factors are defined and estimated. Related terms: beta, loading, factor model, exposure

Beta as a Vector Component

Difficulty: Beginner

One-Sentence Meaning

Beta is a sensitivity or loading, and multiple betas can form an exposure vector.

Memory Jogger

One beta is one directional loading; many betas form a risk fingerprint.

Why It Matters in Quant Research

Used to describe exposure to market, sectors, styles, macro variables, or custom factors.

Simple Market / Trading Example

A stock might have beta 1.2 to market, 0.4 to momentum, and -0.1 to value.

Formula / Notation Intuition

```
r_i = beta_i * f + epsilon_i ; multi-factor: r_i = B_i f + epsilon_i
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Betas help translate raw returns into interpretable factor relationships.

Python / System Implementation Idea

Estimate rolling betas and compare them to static estimates to monitor exposure drift.

Confusions to Avoid

Beta is estimated, not known. It can change through time and can be unstable in small samples.

Related Terms

regression, factor loading, exposure vector

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

REGRESSION

Regression in Matrix Form

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Linear regression can be written as a matrix equation that maps features to predicted targets through coefficients.

Memory Jogger

Regression is matrix-vector prediction plus error.

Why It Matters in Quant Research

This view is essential for understanding how models scale from one feature to many features.

Simple Market / Trading Example

A model predicting next-day returns from momentum, volatility, and value is X beta plus residuals.

Formula / Notation Intuition

```
y = X beta + epsilon
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Matrix form connects regression, optimization, projection, and statistical diagnostics.

Python / System Implementation Idea

Build X, y, and beta with explicit column names so coefficients remain interpretable.

Confusions to Avoid

The compact formula hides many assumptions: alignment, stationarity, noise behavior, and feature quality.

Related Terms

design matrix, coefficients, residuals, least squares

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

REGRESSION

Least Squares

Quant Research Field Guide

Difficulty: Beginner

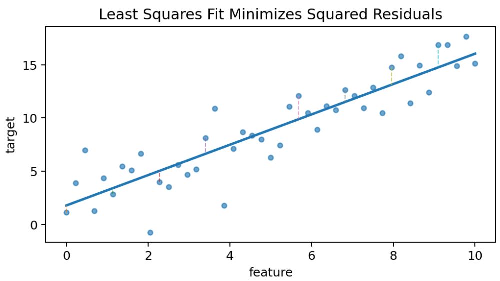
One-Sentence Meaning

Least squares chooses coefficients that minimize the sum of squared residuals.

Memory Jogger

Fit the line by making squared errors as small as possible.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

OLS regression is a baseline modeling tool for returns, factors, hedges, and exposures.

Simple Market / Trading Example

A factor regression estimates the beta values that best explain an asset's return history under a linear model.

Formula / Notation Intuition

```
beta_hat = argmin_beta ||y - X beta||_2^2
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

Least squares is the foundation of many model-fitting techniques. Implementation idea: Always evaluate out-of-sample behavior; a low in-sample residual norm is not enough.

Confusions to Avoid + Related Terms

OLS can fit noise if the feature set is unstable, too large, or poorly validated. Related terms: residuals, projection, regression, normal equations

REGRESSION

Projection View of Regression

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Regression projects the target vector onto the column space of the feature matrix.

Memory Jogger

The model can only explain what lies in its feature space.

Why It Matters in Quant Research

This explains fitted values, residuals, and why adding features can reduce in-sample error.

Simple Market / Trading Example

If y is the target return vector and X contains factors, y_hat is the part of y explained by those factors.

Formula / Notation Intuition

```
y = y_hat + e ; y_hat in Col(X) ; e orthogonal to Col(X)
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Projection gives a geometric interpretation of model fitting.

Python / System Implementation Idea

Inspect residual correlations with features to check whether major linear structure remains.

Confusions to Avoid

A better projection in-sample can still be worse out-of-sample if it captures noise.

Related Terms

column space, residual, orthogonality, least squares

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

REGRESSION

Residual Vectors

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A residual vector stores the difference between actual values and model predictions.

Memory Jogger

Residuals are what the model did not explain.

Why It Matters in Quant Research

Residuals are used for diagnostics, alpha definitions, error analysis, and factor-neutral strategy construction.

Simple Market / Trading Example

A stock's residual return after removing market and sector effects can be studied as idiosyncratic return.

Formula / Notation Intuition

```
e = y - X beta_hat
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Residual analysis is where many hidden model failures become visible.

Python / System Implementation Idea

Plot residuals through time and check their distribution, autocorrelation, and relationship to omitted variables.

Confusions to Avoid

Residuals are not automatically alpha. They may be noise, missing risk, or bad data.

Related Terms

error vector, fitted values, idiosyncratic return

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

REGRESSION

Multicollinearity

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Multicollinearity occurs when features contain overlapping linear information.

Memory Jogger

Several columns are trying to say the same thing.

Why It Matters in Quant Research

It can make regression coefficients unstable even when predictions look acceptable.

Simple Market / Trading Example

Momentum measured over 60 days and momentum measured over 63 days may be highly collinear.

Formula / Notation Intuition

```
high corr(x_j, x_k) or small singular values indicate overlap
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is a feature-quality and interpretability problem.

Python / System Implementation Idea

Use correlation matrices, VIF-style checks, singular values, or regularization when feature overlap is severe.

Confusions to Avoid

Removing collinearity does not automatically improve prediction, but it often improves stability and interpretation.

Related Terms

correlation, rank, condition number, regularization

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

NUMERICAL LINEAR ALGEBRA

Conditioning and Numerical Stability

Quant Research Field Guide

Difficulty: Beginner

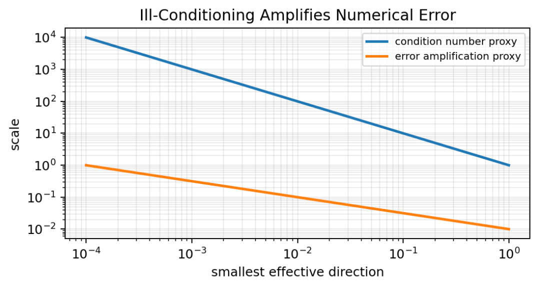
One-Sentence Meaning

Conditioning describes how much small input errors can be amplified by a calculation.

Memory Jogger

A fragile matrix turns tiny errors into big output changes.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Poor conditioning can break optimization, inversion, regression, and covariance-based risk models.

Simple Market / Trading Example

A nearly redundant feature matrix may produce wildly changing coefficients with small data changes.

Formula / Notation Intuition

```
condition_number(A) = largest_singular_value / smallest_singular_value
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

This is the numerical safety layer of linear algebra. Implementation idea: Monitor condition numbers and prefer stable solvers, shrinkage, or regularization when matrices are ill-conditioned.

Confusions to Avoid + Related Terms

A result can be mathematically valid but numerically unreliable. Related terms: condition number, singular values, regularization

Regularization Intuition

Difficulty: Beginner

One-Sentence Meaning

Regularization stabilizes a model by penalizing overly large or unstable parameter values.

Memory Jogger

Pay a penalty to avoid fragile fits.

Why It Matters in Quant Research

Used in ridge regression, lasso, covariance shrinkage, and portfolio optimization constraints.

Simple Market / Trading Example

Ridge regression adds a penalty to discourage extreme coefficients when features overlap.

Formula / Notation Intuition

```
ridge objective = ||y - X beta||_2^2 + lambda ||beta||_2^2
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Regularization is a practical response to noisy high-dimensional data.

Python / System Implementation Idea

Tune regularization using out-of-sample or walk-forward validation, not just in-sample fit.

Confusions to Avoid

Too little regularization overfits; too much underfits. The penalty must be validated.

Related Terms

ridge, lasso, shrinkage, overfitting

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

DECOMPOSITIONS

PCA: Principal Component Analysis

Quant Research Field Guide

Difficulty: Beginner

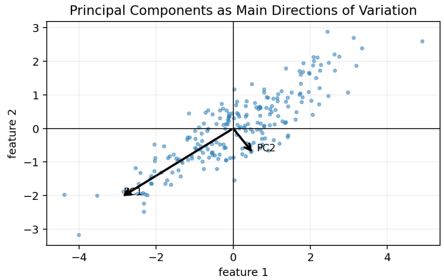
One-Sentence Meaning

PCA finds orthogonal directions that explain the most variance in the data.

Memory Jogger

Rotate the coordinate system toward the largest variation.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Used for risk compression, feature reduction, curve modeling, and understanding common movement in asset returns.

Simple Market / Trading Example

In yield-curve analysis, the first components often resemble level, slope, and curvature movements.

Formula / Notation Intuition

```
components = eigenvectors of the covariance matrix
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

PCA is a dimensionality-reduction and diagnostic tool. Implementation idea: Standardize inputs if variable scale should not dominate PCA.

Confusions to Avoid + Related Terms

PCA explains variance, not necessarily tradable edge or economic causality. Related terms: eigenvectors, covariance matrix, principal components

DECOMPOSITIONS

SVD: Singular Value Decomposition

Quant Research Field Guide

Difficulty: Beginner

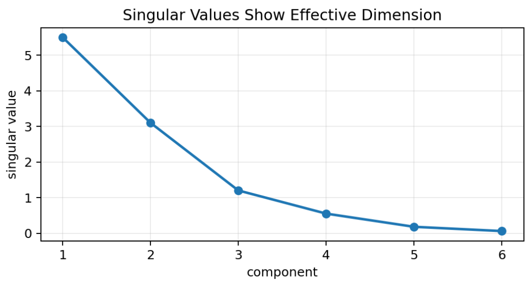
One-Sentence Meaning

SVD decomposes a matrix into left directions, singular values, and right directions.

Memory Jogger

Break a matrix into its strongest patterns.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

SVD supports PCA, dimensionality reduction, matrix rank diagnostics, and numerical stability checks.

Simple Market / Trading Example

A feature matrix with a steep singular-value drop may have only a few strong effective directions.

Formula / Notation Intuition

```
A = U S V^T
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

SVD is one of the most important tools in numerical linear algebra. Implementation idea: Use singular values to diagnose effective dimension and feature redundancy.

Confusions to Avoid + Related Terms

The decomposition is powerful, but the components still need interpretation and stability checks. Related terms: rank, PCA, singular values, conditioning

DECOMPOSITIONS

Dimensionality Reduction

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Dimensionality reduction represents high-dimensional data with fewer variables while trying to preserve important structure.

Memory Jogger

Compress without losing what matters.

Why It Matters in Quant Research

Used to reduce noise, stabilize models, visualize regimes, and control overfitting.

Simple Market / Trading Example

A 50-feature model might be compressed into 5 principal components before regression.

Formula / Notation Intuition

`Z = X W where W maps original features to lower-dimensional components`

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is part of the feature-control layer of research.

Python / System Implementation Idea

Compare model performance and stability before and after dimensionality reduction.

Confusions to Avoid

Compression can remove useful signal if the retained directions are chosen poorly.

Related Terms

PCA, SVD, feature selection, manifold learning

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

RESEARCH SYSTEMS

Feature Matrix for Machine Learning

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A feature matrix stores observations as rows and model inputs as columns.

Memory Jogger

The design matrix is the model's input table.

Why It Matters in Quant Research

Every ML workflow needs a clean feature matrix and target vector before fitting anything.

Simple Market / Trading Example

For daily stock prediction, rows might be stock-date observations and columns might be lagged returns, volatility, volume, and factor scores.

Formula / Notation Intuition

```
model.fit(X_train, y_train)
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is where raw market data becomes model-ready numerical structure.

Python / System Implementation Idea

Build feature pipelines that preserve timestamps and explicitly prevent future data from entering past rows.

Confusions to Avoid

Feature matrices are easy to corrupt with lookahead bias, misalignment, or survivorship bias.

Related Terms

X matrix, target y, supervised learning, design matrix

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

RESEARCH SYSTEMS

Rolling Windows as Matrices

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A rolling window turns a moving slice of time-series data into repeated matrices for estimation and testing.

Memory Jogger

Each window is a temporary matrix view of recent history.

Why It Matters in Quant Research

Used for rolling covariance, rolling regression, walk-forward testing, and adaptive feature estimates.

Simple Market / Trading Example

A 252-day rolling return matrix can estimate covariance for today's portfolio risk forecast.

Formula / Notation Intuition

```
for each t: X_window = X[t-L:t, :]
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Rolling matrices connect time-series validation to linear algebra objects.

Python / System Implementation Idea

Store window start and end times so every estimate is auditable and free of lookahead bias.

Confusions to Avoid

Window length controls a tradeoff between responsiveness and estimation noise.

Related Terms

rolling covariance, walk-forward, estimation window

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

RESEARCH SYSTEMS

Gram and Similarity Matrices

Quant Research Field Guide

Difficulty: Beginner

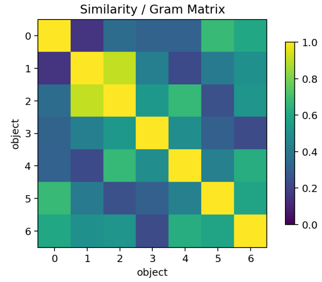
One-Sentence Meaning

A Gram matrix stores pairwise dot products or similarities between objects.

Memory Jogger

It is a table of how every object relates to every other object.

Visual Block



Synthetic visual generated for this manual. Use it for intuition and structure, not exact market measurement.

Why It Matters in Quant Research

Used in kernel methods, signal similarity, clustering, and redundancy diagnostics.

Simple Market / Trading Example

A signal similarity matrix can reveal that several strategies are effectively the same return stream.

Formula / Notation Intuition

```
G = X X^T or G_ij = x_i dot x_j
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Research-System Fit + Implementation

This is a structural map of relationships inside a research library. Implementation idea: Use heatmaps to inspect similarity matrices visually before relying on summary metrics.

Confusions to Avoid + Related Terms

High similarity does not automatically mean identical economic logic, but it does deserve investigation. Related terms: dot product, cosine similarity, kernel matrix, clustering

RESEARCH SYSTEMS

Missing Data and Alignment

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Financial matrices often contain missing values or mismatched timestamps, and alignment controls whether calculations are meaningful.

Memory Jogger

Bad alignment quietly poisons good math.

Why It Matters in Quant Research

Price histories, fundamentals, delistings, holidays, and asset universes rarely line up perfectly.

Simple Market / Trading Example

If one asset's return is shifted by one day, covariance and model estimates can become invalid.

Formula / Notation Intuition

```
valid_matrix = aligned_data.dropna() is simple but not always unbiased
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

Data alignment is the operational layer beneath every matrix calculation.

Python / System Implementation Idea

Audit missingness patterns before deciding whether to drop, forward-fill, backfill, or model missing values.

Confusions to Avoid

Dropping all missing rows may create survivorship or selection bias; filling missing values can create artificial patterns.

Related Terms

NaN handling, index alignment, survivorship bias

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

RESEARCH SYSTEMS

Broadcasting and Shape Bugs

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

Broadcasting lets arrays with different shapes interact automatically, which is powerful but dangerous.

Memory Jogger

The code may run while the math is wrong.

Why It Matters in Quant Research

Many portfolio and ML bugs come from unintended broadcasting across dates, assets, or features.

Simple Market / Trading Example

A vector of asset means might accidentally be subtracted across the wrong axis if dimensions are not explicit.

Formula / Notation Intuition

`A + b can broadcast b across rows or columns depending on shape`

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This is one of the most practical debugging topics in numerical research code.

Python / System Implementation Idea

Use explicit reshaping and assert statements for expected dimensions before critical operations.

Confusions to Avoid

No error message does not mean no bug. Broadcasting can silently create a plausible-looking matrix.

Related Terms

shape, axis, vectorization, NumPy

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

RESEARCH SYSTEMS

Linear Algebra Checklist for Backtests

Quant Research Field Guide

Difficulty: Beginner

One-Sentence Meaning

A backtest should verify the shapes, labels, and meanings of all vectors and matrices before computing performance.

Memory Jogger

Most bad math starts before the formula.

Why It Matters in Quant Research

Backtests combine return matrices, signal matrices, weight vectors, cost matrices, and risk matrices.

Simple Market / Trading Example

A signal matrix must align with future returns correctly: today’s signal should not use tomorrow’s return information.

Formula / Notation Intuition

```
portfolio_returns_t = sum_i weights_{t-1,i} * returns_{t,i}
```

Read the formula as the compact expression of the page. Then connect it back to vectors, matrices, weights, features, or risk objects.

Where This Fits in the Research System

This page turns linear algebra into practical research hygiene.

Python / System Implementation Idea

Add checks for date alignment, asset alignment, no future leakage, shape compatibility, and finite values.

Confusions to Avoid

A backtest can be mathematically fast and still logically invalid if data alignment is wrong.

Related Terms

lookahead bias, alignment, weights, return matrix

Quick Recall / Notebook Drill

- State the object type and expected shape before using it.
- Create a tiny numerical example and compute it by hand once.
- Explain how a label, timestamp, or component-order mistake would break this concept in a backtest.

Worked Calculation - Dot Product Portfolio Return

Difficulty: Beginner

Setup

Weights are w = [0.50, 0.30, 0.20]. Asset returns are r = [0.02, -0.01, 0.04]. Compute portfolio return with w^T r.

Step-by-Step

```
w^T r = 0.50*0.02 + 0.30*(-0.01) + 0.20*0.04 = 0.010 - 0.003 + 0.008 = 0.015
```

The portfolio return is 1.5%. The dot product compresses matching asset returns and weights into one scalar portfolio return.

Checks

- Weights and returns must refer to the same assets in the same order.
- If weights are from yesterday and returns are from today, that timing must be intentional.
- Gross exposure and net exposure are separate diagnostics from return itself.

Worked Calculation - Matrix-Vector Prediction

Difficulty: Beginner

Setup

Let X have 3 observations and 2 features. X = [[1, 0.5], [1, -0.2], [1, 1.1]]. Let beta = [0.01, 0.03]. Compute y_hat = X beta.

Step-by-Step

```
obs1: 1*0.01 + 0.5*0.03 = 0.025 obs2: 1*0.01 + (-0.2)*0.03 = 0.004 obs3: 1*0.01 + 1.1*0.03 = 0.043
```

Each row creates one prediction. Matrix-vector multiplication is repeated dot products.

Interpretation

The first column can represent an intercept and the second column a feature. The beta vector gives the fitted effect of each column.

Implementation Note

In Python: y_hat = X @ beta. Confirm X.shape is (3,2) and beta.shape is (2,) or (2,1).

REVIEW / APPENDIX

Worked Calculation - Portfolio Variance

Quant Research Field Guide

Difficulty: Beginner

Setup

Let w = [0.60, 0.40]. Let Sigma = [[0.0400, 0.0100], [0.0100, 0.0225]]. Compute variance = w^T Sigma w.

Step-by-Step

```
Sigma w = [0.0400*0.60 + 0.0100*0.40, 0.0100*0.60 + 0.0225*0.40] = [0.0280, 0.0150] w^T Sigma w =
0.60*0.0280 + 0.40*0.0150 = 0.0228
```

The portfolio standard deviation is sqrt(0.0228), about 15.1% if the covariance units are annualized.

Checks

- Covariance units must match the horizon you want to report.
- Weights must match the covariance matrix asset order.
- Off-diagonal covariance terms affect portfolio risk; they are not optional.

REVIEW / APPENDIX

Worked Calculation - Factor Exposure

Quant Research Field Guide

Difficulty: Beginner

Setup

Portfolio weights are w = [0.50, 0.30, 0.20]. Asset factor exposure matrix B has rows as assets and columns as factors: MKT and VALUE.

Example

```
B = [[1.1, 0.2], [0.8, -0.1], [1.3, 0.4]] w^T B = [0.50,0.30,0.20] B = [1.05, 0.15]
```

The portfolio has market exposure 1.05 and value exposure 0.15 under this factor model.

Interpretation

A factor exposure calculation is just a weighted average of asset exposures. The linear algebra keeps many factors organized at once.

Implementation Note

Use exposures = weights.T @ B after verifying that weight labels match B row labels.

Diagnostic Table - Shape and Alignment Bugs

Difficulty: Beginner

Failure Modes

Wrong output shape	The inner dimensions may not match the intended operation, or an accidental transpose changed the meaning.
Code runs but output is wrong	Broadcasting may have applied a vector across the wrong axis.
Portfolio return looks impossible	Weights and returns may be misaligned by asset order or timestamp.
Covariance matrix unstable	Too many assets relative to history, missing data, or highly redundant series.
Regression coefficients jump wildly	Feature matrix may be ill-conditioned or multicollinear.
Signal backtest too good	Feature matrix may contain lookahead leakage or shifted targets.

Debug Checklist

- Print shape and axis meaning after every major transform.
- Check labels before converting DataFrames to raw arrays.
- Assert that dates, assets, and feature columns are aligned.
- Test formulas on a tiny hand-computable example before scaling up.

Diagnostic Table - Linear Model Stability

Difficulty: Beginner

Symptoms and Interpretations

High in-sample fit, poor out-of-sample fit	Likely overfitting, noisy features, or unstable relationships.
Large opposite-signed coefficients	Possible multicollinearity or unstable feature interactions.
Tiny singular values	Feature matrix contains weak or redundant directions.
High condition number	Small data changes may produce large coefficient changes.
PCA components flip or rotate over time	Dominant variation directions may be unstable across regimes.
Covariance optimizer creates extreme weights	Risk matrix may be noisy or nearly singular; constraints and shrinkage may be needed.

System Rule

Stability matters more than algebraic elegance. A formula that is correct but unstable can still destroy a research workflow.

Formula Summary Sheet I

Difficulty: Beginner

Vector Basics

`x + y = [x1+y1, ..., xn+yn] c x = [c*x1, ..., c*xn] x dot y = sum_i x_i y_i ||x||_2 = sqrt(sum_i x_i^2)`

Core vector operations used in returns, weights, features, and exposures.

Matrix Basics

`(m x n) times (n x p) = (m x p) y_hat = X beta A^T flips rows and columns A^{-1} A = I when inverse exists`

Shape rules and basic matrix operations.

Linear Systems

`A x = b rank(A) = number of independent columns Null(A) = {x : A x = 0} Col(A) = span of columns of A`

Structure behind constraints, hedges, and model capacity.

REVIEW / APPENDIX

Formula Summary Sheet II

Quant Research Field Guide

Difficulty: Beginner

Portfolio and Risk

`R_p = w^T r Var(R_p) = w^T Sigma w portfolio_factor_exposure = w^T B Sigma_ij = Cov(r_i, r_j)`

Key formulas linking linear algebra directly to portfolio construction.

Regression and Projection

`y = X beta + epsilon beta_hat = argmin_beta ||y - X beta||_2^2 e = y - y_hat y_hat is in Col(X)`

Regression as matrix prediction and geometric projection.

Decompositions

`A v = lambda v A = U S V^T condition_number(A) = largest_singular_value / smallest_singular_value`

Eigenvectors, SVD, and numerical stability diagnostics.

Comparison Sheet - Vector, Matrix, Covariance, Factor Model

Difficulty: Beginner

Object Comparison

Vector	One ordered list. Examples: asset returns, portfolio weights, feature values, factor exposures.
Matrix	Rectangular table. Examples: return history, feature matrix, factor exposure table, covariance matrix.
Dot product	Compresses two aligned vectors into one scalar. Examples: portfolio return, linear score.
Covariance matrix	Stores volatility and co-movement. Used for portfolio risk and optimization.
Factor model	Uses exposure matrices and factor returns to explain asset or portfolio behavior.
Regression matrix	Uses a design matrix X to estimate coefficient vector beta and predictions y_hat.
PCA / SVD	Finds dominant directions and effective dimension in high-dimensional data.

Unifying View

Most quant linear algebra reduces to three acts: represent data as vectors or matrices, transform it with multiplication or decomposition, then interpret the output as return, risk, exposure, signal, or residual.

REVIEW / APPENDIX

Quick Recall Review I

Quant Research Field Guide

Difficulty: Beginner

Questions

- What does the dot product compute in portfolio-return language?
- Why must vector components be aligned before addition or dot products?
- What does the shape T x N usually mean for a return matrix?
- What is the difference between covariance and correlation?
- What does rank tell you about a feature matrix?

Mini Tasks

- Write a 3-asset weight vector and a 3-asset return vector, then compute the dot product.
- Create a 4 x 2 feature matrix and a 2-element beta vector, then compute predictions.
- Explain in words why covariance terms matter for portfolio variance.
- Describe one way broadcasting can create a silent bug.

Quick Recall Review II

Difficulty: Beginner

Questions

- What does it mean for two vectors to be orthogonal?
- Why is the null space useful for factor-neutral portfolios?
- What is multicollinearity and why does it destabilize coefficients?
- What does PCA try to preserve?
- Why should you prefer solving A x = b directly instead of computing an inverse in production code?

System Checks

- Check row and column meaning before every matrix multiplication.
- Check that feature columns in training and live prediction match exactly.
- Check covariance matrix eigenvalues or singular values before optimization.
- Check residuals after regression instead of trusting coefficients alone.

REVIEW / APPENDIX

Final Concept Map and Index

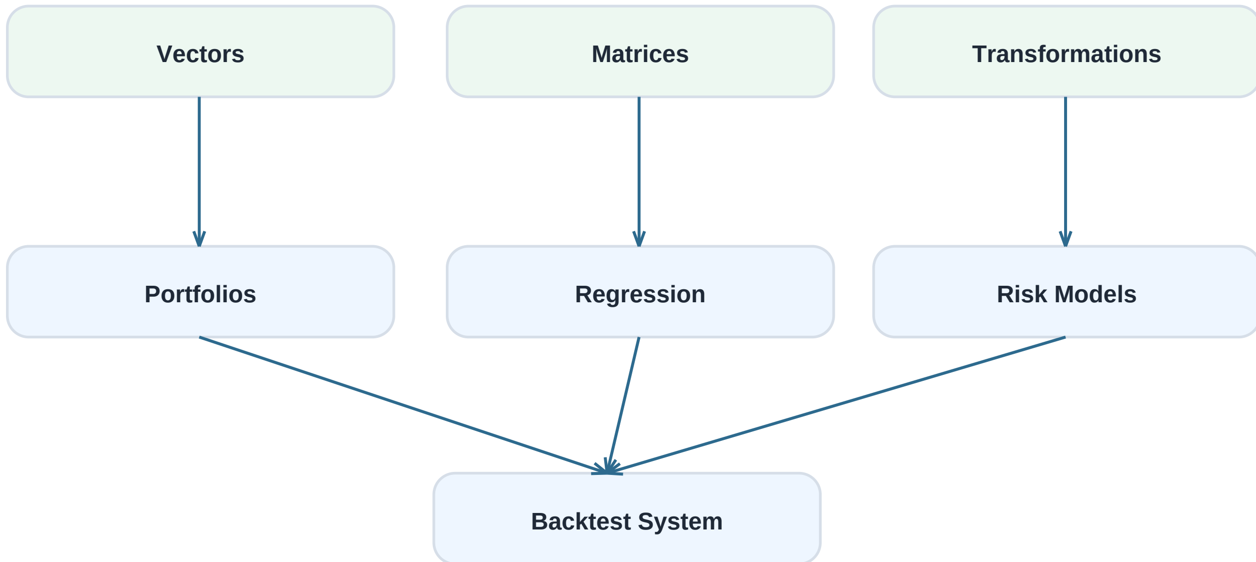
Quant Research Field Guide

Difficulty: Beginner

Big Picture Summary

Linear algebra gives quant research a compact way to represent many assets, many features, many observations, and many constraints. Vectors describe states and decisions. Matrices describe datasets and transformations. Dot products produce scores and portfolio returns. Covariance matrices describe risk. Regression and decompositions turn data into model structure and diagnostics.

Concept Map



Related-Term Index

- array, axis, basis, beta, broadcasting, coefficient vector, column space, condition number, correlation matrix, covariance matrix, design matrix, determinant, dimension, dot product, eigenvalue, eigenvector, exposure matrix, factor model, feature matrix, Gram matrix, identity matrix, inverse matrix, least squares, linear combination, linear transformation, matrix multiplication, multicollinearity, norm, null space, orthogonality, PCA, portfolio variance, projection, rank, residual vector, singular value, SVD, transpose, weight vector.